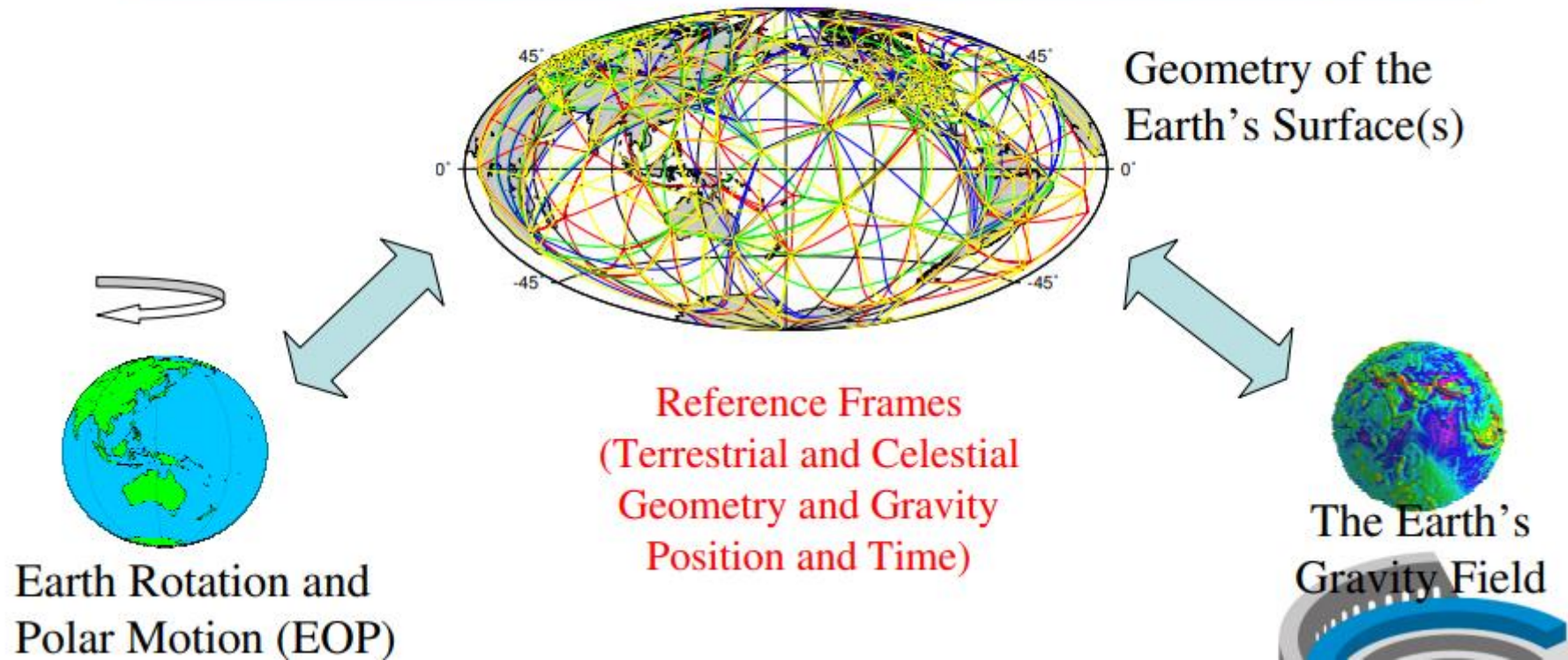


The Classical “Pillars” of Geodesy



Modern Geodesy

Geodesy now defined in terms of the following *capabilities*:

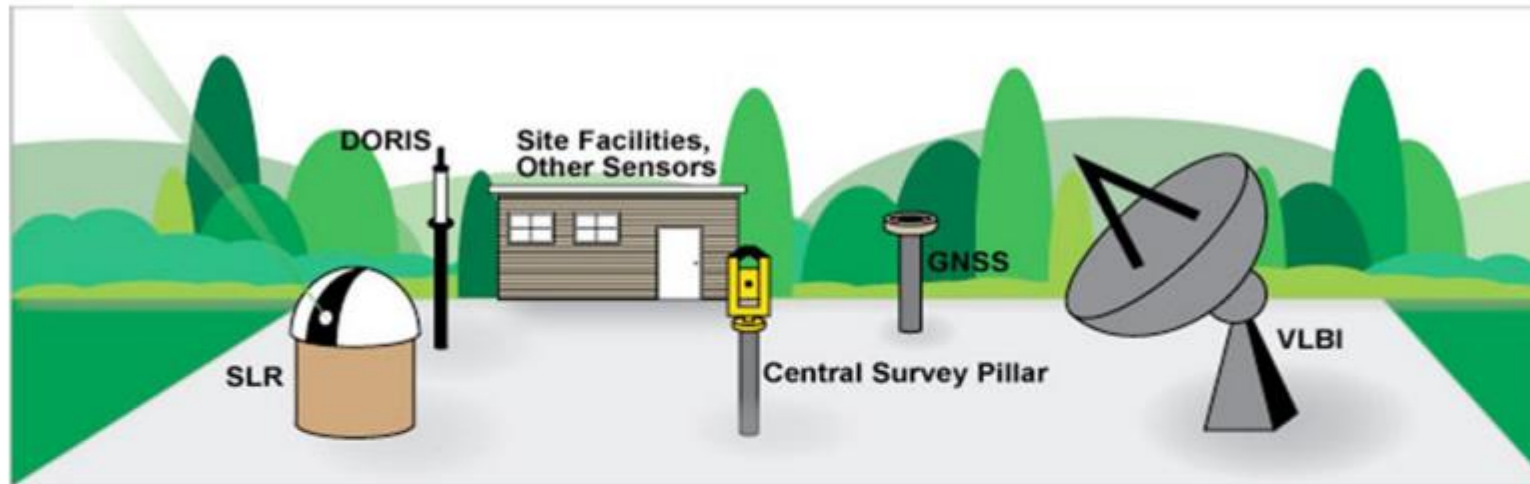
1. Determination of precise global, regional & local 3-D (static or kinematic) *positions on or above the Earth's surface*.
2. Mapping of *land, sea & ice surface geometry*.
3. Determination of the Earth's (time & spatially) *variable gravity field*.
4. Measurement of *dynamical phenomena*:
 - Solid Earth (incl. cryosphere): surface deformation, crustal motion, GIA, polar motion, earth rotation, tides, water cycle, mass transport, etc.
 - Atmosphere: refractive index, T/P/H profiles, TEC, circulation, etc.
 - Ocean: sea level, sea state, circulation, etc.



Space Geodesy

- Measurement of precise Coordinates for extra galactic Radio sources (Quasars) and their structure
- Determination of precise global, regional and local 3D-coordinates
- Determination of the Rotation rate of the Earth to allow the transformation between the two frames of reference
- Determination of the Earth gravity field and its variation over time

GGOS – Global Geodetic Observing System



Motivation – why learn about time?

- Because we are working with different time scales depending on the applications
- Because our satellite geodetic observation methods can provide measures of physical changes with a very high precision
- Because with many observation methods precise time synchronization or time stamping is important
- Because the transmission time of a bit of a GPS satellite signal is approximately 0.078 seconds. To determine a position with GPS we need to know the time very well

System Earth – Relevant Timescales

Lithosphere:

Plate Tectonics ↔ Earthquakes

Millions of Years ↔ several Seconds
cm/year ↔ km/s

Hydrosphere:

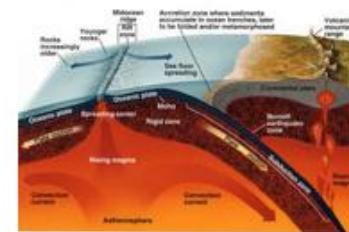
Sea Level Rise ↔ Tsunami

3 mm/year ↔ 300 m/s

Atmosphere:

Climate ↔ Weather

years – decades ↔ hours – days

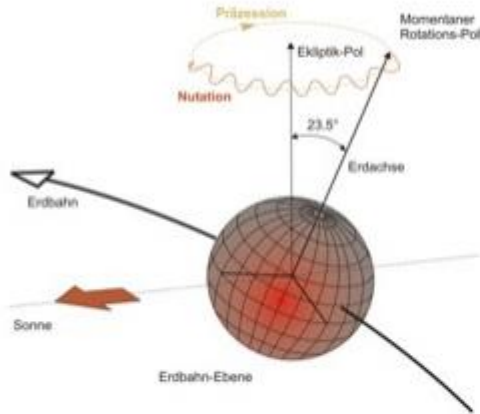


→ Requirements

Measurement techniques of extremely high resolution and stability

Quantification of very small and slow processes vs. highly dynamic realtime

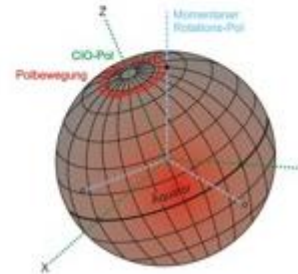
Earth Rotation links ITRF to ICRF



a) gravitational attraction of sun and moon on a near spherical object give rise to precession and nutation

b) the rotation rate of the Earth is not constant. Deceleration by dissipation and variation by momentum exchange. Free oscillations excited by ocean, atmosphere

c) mass redistribution on Earth and the fact that the figure axis and the axis of Inertia are not coinciding, give rise to polar motion



Solar time

- Defined by Earth rotation with respect to the sun
-> one rotation is 24 hours
- Since Earth rotation is not constant neither is the solar time reference. The length of a day varies with up to 16 minutes during a year
- In practise, a mean time is used:
- Greenwich Mean Time – also referred to as GMT or ZULU

Greenwich Apparent Sideral Time (GAST)

- GAST is defined as the angle between the reference meridian (Greenwich) and the X-axis of the CIS (i.e. the direction to the vernal equinox)
- GAST provides the current mean sideral time at the reference meridian
- GAST is defined as an angle, but related to time by considering the Earth rotation rate given as an angular velocity
 - Earth rotation within an hour is approx. 15°
 - The width of time zones are approx. 15° in longitude
- GAST indicates the rotation, about the Z-axis of the inertial coordinates system, of the ECEF coordinate system

Sidereal time

- Defined by Earth rotation with respect to distant stars
- 1 sidereal day (SID) = 23 hours and 56 minutes in solar time
 - or 1 solar day = 1.003 sidereal day
- Since Earth rotation is not constant neither is the sidereal time reference.
- In practise, a mean time is used: Mean sidereal time
- Interesting fact: The GPS constellation repeats itself after half a SID

Atomic time (1)

- The current official SI-definition of the atomic time scale is since 1967 determined by cesium
 - 1 second = the time it takes for 9,192,631,770 periods of radiation of the cesium-133 atom
- Atomic time is homogeneous and continuous (no jumps) and therefore attractive for many satellite geodetic applications
- The fundamental atomic time scale is called *International Atomic Time* (abbreviated TAI)

Atomic time (2)

- Atomic time is not referenced to Earth rotation and over time it therefore drifts further and further away from solar time
- This has led to the introduction of *Universal Time Coordinated*, UTC which is the official time used in many countries today
- The definition of TAI and UTC is controlled by the “*Bureau International des Poids et Mesures*” (BIPM) in Paris
<http://www1.bipm.org>

GPS time (1)

- For GPS-positioning it is crucial with a homogeneous *and* continuous time scale. Therefore a new time scale was defined to be used for GPS, called *GPS time*
- GPS time is defined as UTC but without the leap seconds
 - GPS time = UTC time at January 6th, 1980, @ 00:00 hours
 - GPS time in 2020 equals UTC + 18 seconds + a few nanoseconds
- GPS time is maintained by the US Naval Observatory
<http://www.navcen.uscg.gov>

GPS, UTC, and TAI Clocks

Universal Time Coordinated (UTC)

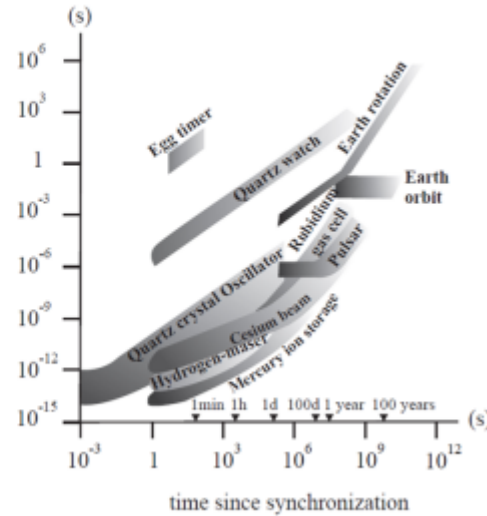
- *Coordinated Universal Time* (UTC) is defined as:
 - 1 second UTC = 1 second TAI
- UTC time is corrected for changes in the rotation of the Earth by an integer number of seconds (leap seconds)
- Leap seconds are introduced, when considered necessary, on either June 30th or December 31st
 - Most recent leap second introduced Dec. 31st 2016 (previous was in June 2015)
- Current difference: TAI – UTC = 37 seconds
- UTC is homogeneous, but not continuous because of the leap seconds

GPS time (2)

- A time epoch given in GPS time is referenced as:
 - Week number, counted since January 6th, 1980
 - Number of seconds within the week (counter is reset at midnight between Saturday and Sunday, i.e. from 604800 to 0 seconds)
- Current GPS week is: 2120
- For some applications, GPS time is also given as:
 - Week number, day of week, seconds of day
- In Assignment 2 we will work with GPS time

Stability of clocks and frequency standards

- A precise and well defined time reference, only makes sense if we have clocks sufficiently good to “keep the time”
- For example for GPS:
An error of 1 microsecond in the GPS receiver clock induces an error of 300 meter in the range estimate to the satellite
- Estimation and modelling of both satellite and receiver clock errors is therefore important for geodetic applications
- Figure: G. Seeber, “Satellite Geodesy”, 2003



GPS satellite clock error (or instability)

- GPS satellites are equipped with two (or three) rubidium and two cesium atomic clocks that are monitored by the GPS control stations
- The current size of the satellite clock errors are between 2 and 750 microseconds
- Satellite clock errors are modelled by polynomials of second order. Coefficients are transmitted to GPS users via the navigation message from the satellites